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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Freelancer ProjectManagementandPaymentTracking System

Submitted by

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Under the Guidance of

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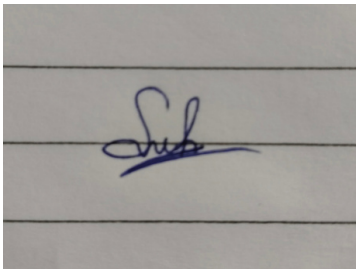
DECLARATION

I hereby declare that the project entitled “Freelancer Project Management and Payment Tracking System” submitted in partial fulfillment of the requirements for the course [Java Programming] is a record of my original work carried out under the guidance of [Aravind], Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for

the award of any
degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.



April 2026

Coimbatore

(Suhas R C)

CERTIFICATE

This is to certify that the project entitled “Freelancer Project Management and Payment Tracking System” is a bona fide work carried out by [SUHAS R.C], [I- “B”-2025], [Bsc.CS], KG College of Arts & Science, in partial fulfillment of the requirements for the course [Java Programming].

This project has been completed under my supervision and guidance during the academic period April 2026.

Place: Coimbatore

Date: 10/04/2026

Submitted for the Viva-Voce Examination held on 10/04/2026

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ABSTRACT

This system provides a simple and effective solution by allowing users to store project details, update payment information, and view pending payments in an organized manner.

The application uses file handling to store data securely, ensuring that project information can be accessed anytime. It reduces manual effort, minimizes errors, and improves productivity. The system is user-friendly and suitable for individual freelancers who need a basic yet reliable tool for managing their work.

The system is user-friendly and suitable for individual freelancers. It can be further enhanced with advanced features such as graphical user interface and database integration

1. INTRODUCTION

Freelancing has become one of the fastest-growing career options in today's digital era. Many individuals prefer working independently rather than being employed in a traditional office environment.

Freelancers take up multiple projects from different clients and work according to their skills and availability. While freelancing offers flexibility and independence, it also brings challenges in managing projects and tracking payments.

Managing multiple projects manually can be difficult and confusing. Freelancers often use notebooks or spreadsheets to maintain records, which may lead to errors, data loss, or missed payments. As the number of projects increases, it becomes harder to keep track of all the details.

To overcome these problems, a computerized system is necessary. The Freelancer Project Management and Payment Tracking System is designed to help freelancers manage their work efficiently. It allows users to store project details, update payment status, and view pending payments in an organized manner.

This system improves accuracy, reduces manual effort, and ensures better productivity. It is simple, user-friendly, and suitable for individual freelancers.

1.1 Objective

The main objective of this project is to develop a system that helps freelancers manage their projects and track payments efficiently. The system aims to reduce manual work and improve accuracy in maintaining records.

Other objectives include:

To store project details systematically

To track payment status (Paid or Pending)

To provide quick access to project information

To reduce errors in manual record keeping

1.2 Overview

The Freelancer Project Management and Payment Tracking System is a software application developed to assis freelancers in managing their projects and tracking payments efficiently. In today's digital world, freelancing ha become a popular career option where individuals work independently for multiple clients. Managing severa projects at the same time can be challenging, especially when it comes to maintaining records and trackin payment status.

This system provides a simple and organized solution to handle such tasks. It allows users to store important project details such as project ID, client name, project title, and payment amount. Each project is recorded along with its payment status, which can be either "Pending" or "Paid." This helps freelancers easily identify which payments are yet to be received.

The application is developed using Java programming language and follows a console-based approach. It uses file handling techniques to store and retrieve data. This ensures that all project information is saved permanently and can be accessed even after the program is closed and reopened. The system is designed in a user-friendly manner so that even users with basic computer knowledge can operate it without difficulty.

The system consists of different modules such as adding a new project, viewing all projects, updating payment status, and displaying pending payments. Each module performs a specific function, making the system well-structured and easy to maintain. The modular design also allows future improvements to be added easily.

Overall, this system helps freelancers improve their productivity by reducing manual work and minimizing errors. It provides a reliable and efficient way to manage freelance activities and ensures better organization of project data.

1.3 Features

The Freelancer Project Management System provides several useful features that help in managing freelance work efficiently. One of the main features is the ability to add and store project details such as project ID, client name, project title, and payment amount. This helps in maintaining a proper record of all projects.

Another important feature is payment tracking. The system allows users to mark payments as “Pending” or “Paid,” making it easy to identify unpaid projects. This reduces confusion and helps freelancers manage their finances better.

The system also provides a feature to view all stored projects. Users can easily access and review project details whenever required. Additionally, it includes an option to display only pending payments, which helps in quick identification of dues.

The application is simple and user-friendly, making it suitable for all users. It uses file handling to store data, ensuring that information is saved permanently and can be accessed anytime

1. **PROGRAM ANALYSIS**

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

Freelancers face many challenges in managing their work, especially when handling multiple projects at the same time.

One of the major problems is maintaining project records manually. Many freelancers use notebooks or spreadsheets, which can lead to errors, data loss, and confusion.

Tracking payments is another significant issue. It becomes difficult to remember which clients have paid and which payments are still pending.
This can result in financial loss and poor management of work.

The lack of a proper system makes it hard to organize project data efficiently. As the number of projects increases, the complexity of managing them also increases. Therefore, there is a need for a system that can simplify project management and payment tracking.

2.2 Project Overview

The Freelancer Project Management and Payment Tracking System is designed to provide a simple solution to the problems faced by freelancers. It allows users to store project details, update payment status, and view pending payments in an organized manner.

The system is developed using Java programming language and uses file handling for data storage.

It provides a structured way to manage freelance work and ensures that all information is stored securely.

The application is divided into different modules, each performing a specific function. This makes the system easy to use and maintain. Overall, the project aims to improve efficiency and reduce manual effort.

2.3 Scopes and Limitations

Scope

The scope of this project includes managing freelance projects and tracking payments efficiently. It is suitable for individual freelancers who need a simple system to organize their work.

The system can handle multiple projects and store all necessary details.

However, the system has certain limitations. It does not include a graphical user interface, which may make it less attractive to some users.

It also uses file storage instead of a database, which limits scalability. Additionally, advanced features such as online access and multi-user support are not included.

Limitations:

No graphical user interface (GUI)

Uses file storage instead of database

Limited to basic functionalities

Despite these limitations, the system serves its purpose effectively for basic project management.

1. **SYSTEM DESIGN**

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The system is divided into different modules to perform specific tasks efficiently. The Add Project module allows users to enter and store new project details. The View Projects module displays all stored projects.

The Update Payment module is used to change the payment status of a project from pending to paid. The View Pending Payments module helps users identify unpaid projects easily. Finally, the Exit module allows users to close the application safely.

1. Add Project Module

This module is used to store new project details into the system. The user enters information such as project ID, client name, project name, and amount. By default, the payment status is set to Pending. The entered data is then saved into a file for future reference.

Function:

Accept user input

Create new project record

Store data in file

2. View Projects Module

This module displays all the projects stored in the system. It retrieves data from the file and shows complete details such as project ID, client name, project name, amount, and payment status.

Function:

Read data from file

Display all project records

3. Update Payment Status Module

This module allows the user to update the payment status of a project. The user enters the project ID, and if the project is found, the payment status is changed from Pending to Paid. The updated data is saved back to the file.

Function:

Search project by ID

Update payment status

Rewrite updated data into file

4. View Pending Payments Module

This module helps the user to identify unpaid projects. It filters and displays only those projects whose payment status is Pending.

Function:

Read all records

Filter pending payments

Display unpaid projects

5. Exit Module

This module is used to terminate the program. When the user selects this option, the system stops execution safely.

Function:

Close the application

This module allows the user to exit the application safely after completing the billing process.

When the user selects the exit option, the system terminates the program.

1. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Freelancer Project Management System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis

• System Design

• Program Development

• Testing and Validation

• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of restaurant operations.

Problem analysis involves studying the issues faced by freelancers in managing their work. Manual methods of record-keeping are time-consuming and prone to errors.

Freelancers often forget important details or lose track of payments.

This system analyzes these problems and provides an automated solution. By using a computerized system, data can be stored securely and accessed easily. The system helps in reducing errors and improving efficiency.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

System design defines how the system is structured and how it operates. The Freelancer Management System is designed using a modular approach, where each function is divided into separate modules.

The design ensures that the system is simple, efficient, and easy to maintain. File handling is used for data storage, and proper data flow is maintained between input and output.

The modular design also allows future enhancements to be added easily without affecting the existing system.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

It follows object-oriented programming concepts such as classes and objects. Each project is represented as an object with attributes like ID, client name, project name, amount, and payment status.

File handling techniques are used to store and retrieve data. The program is designed to handle user inputs and perform operations accordingly. Proper coding practices are followed to ensure readability and efficiency.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

The system is tested using various inputs to ensure correct functionality. Different test cases are used, such as adding new projects, viewing records, updating payment status, and checking pending payments.

The results are verified to ensure that the system works accurately. Any errors found during testing are corrected to improve performance.

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates the final bill based on the selected food items and quantities.

The system produces accurate results and performs all required operations successfully. It stores project details, updates payment status, and displays pending payments correctly.

Proper documentation is maintained to explain the working of the system. This helps in understanding the project and making future improvements.The results are demonstrated through program execution.

4.6. Algorithm

Step 1: Start

Step 2: Display menu

Step 3: Read user choice

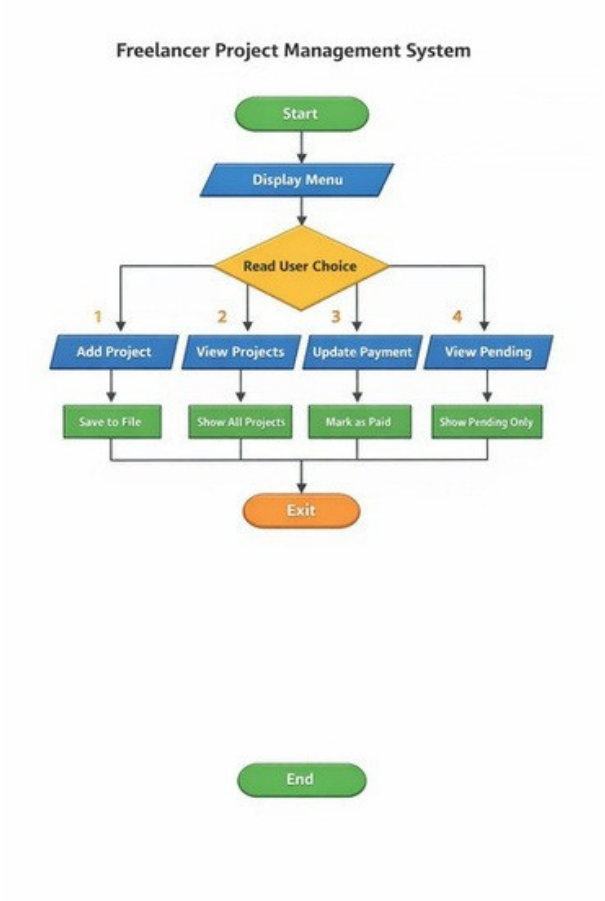
Step 4: Perform selected operation

Step 5: Repeat until exit

Step 6: Stop

4.7. Flowchart

The flowchart represents the working process of the system. It includes start, input, decision-making, process, and end steps.



1. SOURCE CODE

```
import java.io.*;
import java.util.*;

class Project implements Serializable {
    int id;
 String clientName;
 String projectName;
 double amount;
 String paymentStatus;

    public Project(int id, String clientName, String projectName, double amount, String paymentStatus) {
        this.id = id;
        this.clientName = clientName;
        this.projectName = projectName;
        this.amount = amount;
        this.paymentStatus = paymentStatus;
    }

    public void display() {
        System.out.println("ID: " + id);
        System.out.println("Client: " + clientName);
        System.out.println("Project: " + projectName);
        System.out.println("Amount: ₹" + amount);
        System.out.println("Status: " + paymentStatus);
        System.out.println("---------------------------");
    }
```

```
}

public class FreelancerSystem {
 static final String FILE_NAME = "projects.dat";

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        while (true) {
            System.out.println("\n=== Freelancer Management System ===");
            System.out.println("1. Add Project");
            System.out.println("2. View Projects");
            System.out.println("3. Update Payment Status");
            System.out.println("4. View Pending Payments");
            System.out.println("5. Exit");
            System.out.print("Enter choice: ");

            int choice = sc.nextInt();

            switch (choice) {
                case 1:
                    addProject(sc);
                    break;
                case 2:
                    viewProjects();
                    break;
                case 3:
```

```
            updatePayment(sc);
            break;
        case 4:
            viewPending();
            break;
        case 5:
            System.out.println("Exiting...");
            System.exit(0);
        default:
            System.out.println("Invalid choice!");
    }
  }
}

staticvoid addProject(Scanner sc) {
  try{
    System.out.print("Enter ID: ");
    int id = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Client Name: ");
    String client = sc.nextLine();

    System.out.print("Enter Project Name: ");
    String project = sc.nextLine();

    System.out.print("Enter Amount: ");
```

```
        double amount = sc.nextDouble();

        Project p = new Project(id, client, project, amount, "Pending");

        FileOutputStream fos = new FileOutputStream(FILE_NAME, true);
        ObjectOutputStream oos = new ObjectOutputStream(fos) {
            protected void writeStreamHeader() throws IOException {
                reset();
            }
        };

        oos.writeObject(p);
        oos.close();

        System.out.println("Project Added Successfully!");

    }catch (Exception e) {
        System.out.println("Error: " + e);
    }
}

staticvoid viewProjects() {
    try{
        FileInputStream fis = new FileInputStream(FILE_NAME);
        ObjectInputStream ois = new ObjectInputStream(fis);

        while (true) {
```

```
            Project p = (Project) ois.readObject();
            p.display();
        }
    }catch (EOFException e) {
        //End of file
    }catch (Exception e) {
        System.out.println("No records found.");
    }
}

static void updatePayment(Scanner sc) {
    List<Project> list = new ArrayList<>();

    try{
        FileInputStream fis = new FileInputStream(FILE_NAME);
        ObjectInputStream ois = new ObjectInputStream(fis);

        while (true) {
            list.add((Project) ois.readObject());
        }
    }catch (EOFException e) {
    }catch (Exception e) {
    }

    System.out.print("Enter Project ID to update: ");
    int id = sc.nextInt();
```

```
    boolean found = false;

    for(Project p : list) {
        if(p.id == id) {
            p.paymentStatus = "Paid";
            found = true;
            System.out.println("Payment Updated!");
            break;
        }
    }

    if(!found) {
        System.out.println("Project not found!");
        return;
    }

    try{
        ObjectOutputStream oos = new ObjectOutputStream(new FileOutputStream(FILE_NAME));
        for(Project p : list) {
            oos.writeObject(p);
        }
        oos.close();
    }catch (Exception e) {
        System.out.println("Error updating file.");
    }
}
```

```
    staticvoid viewPending() {
        try{
            FileInputStream fis = new FileInputStream(FILE_NAME);
            ObjectInputStream ois = new ObjectInputStream(fis);


            while (true) {
                Project p = (Project) ois.readObject();
                if (p.paymentStatus.equals("Pending")) {
                    p.display();
                }
            }
        } catch (EOFException e) {
        } catch (Exception e) {
            System.out.println("No pending records.");
        }
    }
}
```

1. INPUT AND OUTPUT SCREENS

```
==== Freelancer Project Management System ====

1. Add New Project
2. View All Projects
3. Update Payment Status
4. View Pending Payments
5. Exit

Enter your choice: 1

Enter Project ID: 101
Enter Client Name:  ABC Corp
Enter Project Title: Web Development
Enter Project Amount: 1500
Enter Payment Status (Pending/Paid): Pending

Project added successfully!
Press Enter to continue...
```

```
==== Freelancer Project Management System ====

Pending Payments:

-------------------------------------------------------------
 Project ID   Client Name     Project Title    Amount  Status
-------------------------------------------------------------
   101        ABC Corp      Web Development      1500   Pending
   102        XYZ Ltd       Graphic Design       2000   Pending
-------------------------------------------------------------

Press Enter to continue...
```

1. DISCUSSION & CONCLUSION

7.1 Discussion

TheFreelancer Project Management System provides an efficient solution for managing freelance work. It reduces manual effort and improves accuracy.

The system is simple and easy to use, making it suitable for beginners.

It also helps freelancers keep track of payments and manage their projects effectively. The system can be further enhanced with advanced features

One of the key advantages of this system is its ability to track payment status efficiently. By clearly marking projects as “Paid” or “Pending,” freelancers can easily identify outstanding payments. This reduces the chances

of missing payments and helps in better financial planning. The system also allows users to view all projects at once, which improves transparency and organization.

7.2 Conclusion

The project successfully achieves its objectives by providing a simple and reliable system for managing projects and tracking payments. It improves productivity and reduces errors in manual record-keeping.

The Freelancer Project Management and Payment Tracking System is a useful tool for freelancers. It simplifies project management and payment tracking. The system can be further enhanced to meet advanced requirements. Through this project, the challenges of manual record-keeping have been addressed effectively. The system reduces errors, saves time, and improves the overall productivity of freelancers. It provides a clear and organized structure for managing multiple projects, making it easier to handle work without confusion.

The use of Java programming and file handling techniques ensures that the system is both functional and efficient. The modular design makes the system easy to understand, maintain, and enhance in the future. Although the system has some limitations, such as the absence of a graphical interface and database support, it still serves its purpose effectively for basic project management.

1. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com